

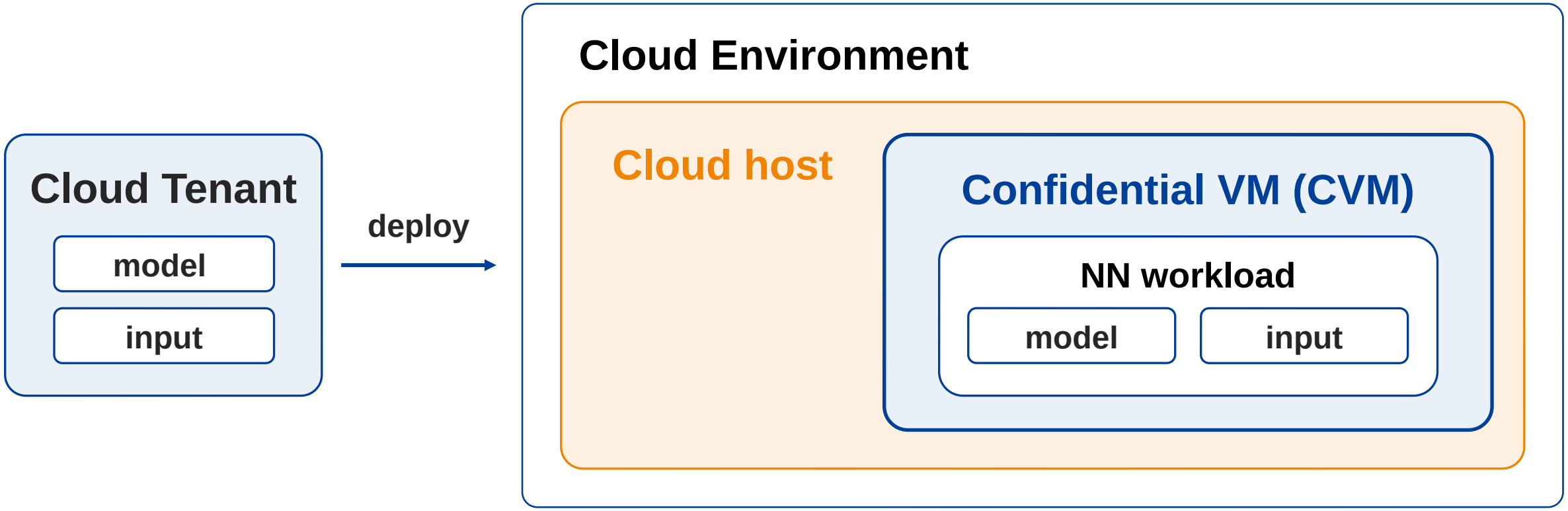


Mitigating Ciphertext Side-Channel Attacks against TEE-Shielded Neural Networks

Qianhui Dai[1], Zhen Huang[1], Guoxing Chen[1], Yan Meng[1], Yinqian Zhang[2], Haojin Zhu[1]

[1]Shanghai Jiao Tong University [2]Southern University of Science and Technology

Cloud-Based NNs Use Confidential Computing



NN workloads in the cloud often run inside CVMs for secure computation

Ciphertext Side Channels in CVMs

Confidential VMs
memory is encrypted

$C = Enc_k(\text{text}, \text{address}, \cancel{\text{counter}})$

One address, over time

Time	Value (hidden)	Ciphertext (visible)
t1	0.00	7f3a…
t2	1.37	9c11…
t3	0.00	7f3a…
t4	0.00	7f3a…
t5	2.05	04e8…

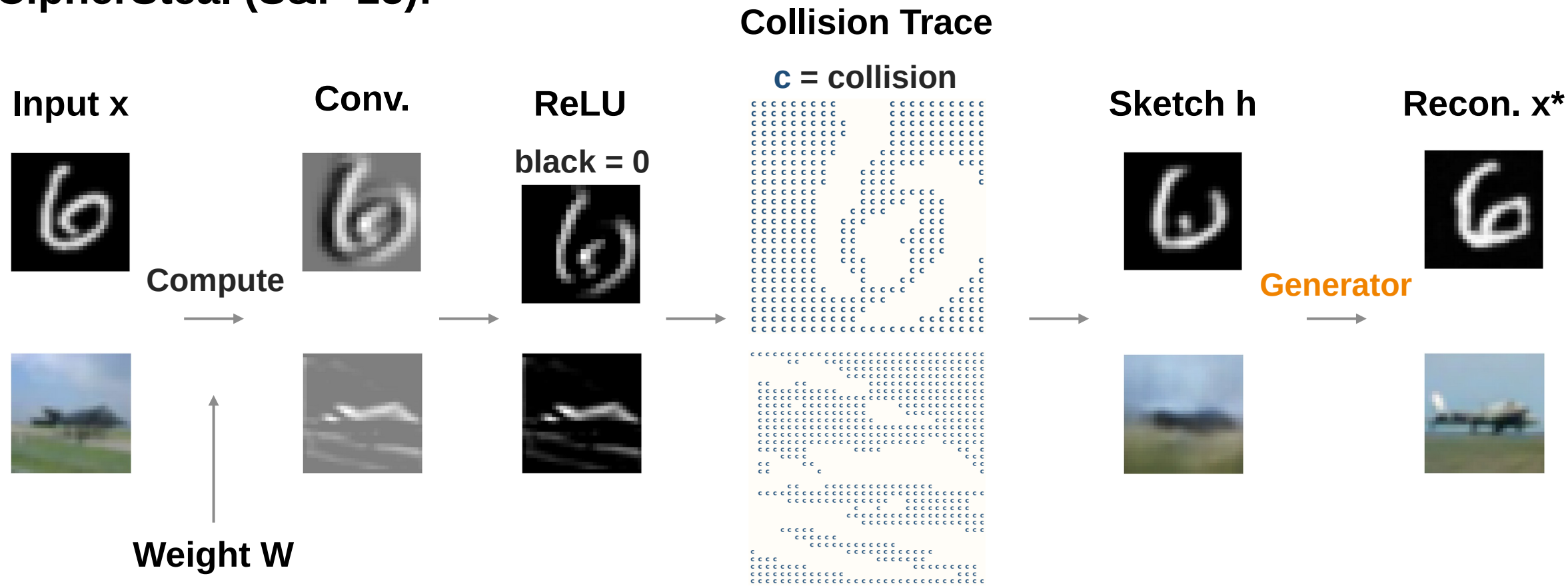
collision

Attacker Capabilities

- ✓ **see ciphertext of every write**
- ✓ know program binary
- ✗ obtain the encryption key
- ✗ modify or replay memory

Ciphertext Side Channels in NNs

CipherSteal (S&P 25):



HyperTheft (CCS 24): the collision trace also leaks the weights W

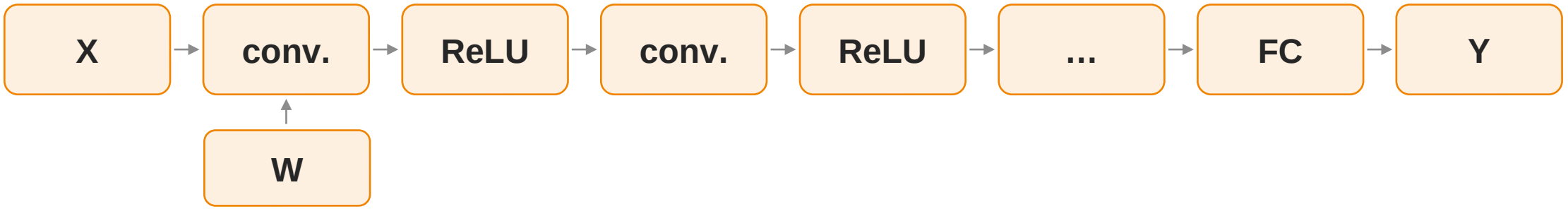
Existing Defenses

CipherFix (USENIX Sec 23)

ZebraFix (RAID 25)



Generic hardening of every write



+692% ~ +2466% inference time

The general defense incurs high overhead

Protection from the Attacker's Perspective

Existing defenses incur high overhead → **Must every collision be protected?**

Goal: no useful victim-specific information from the trace

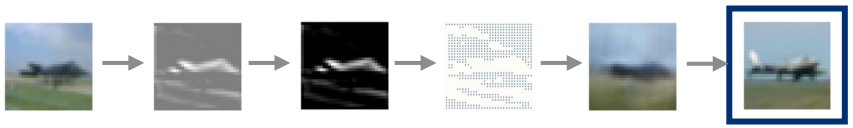
What information does the collision trace reveal?

? →

Assessment limited to attacker-provided outputs

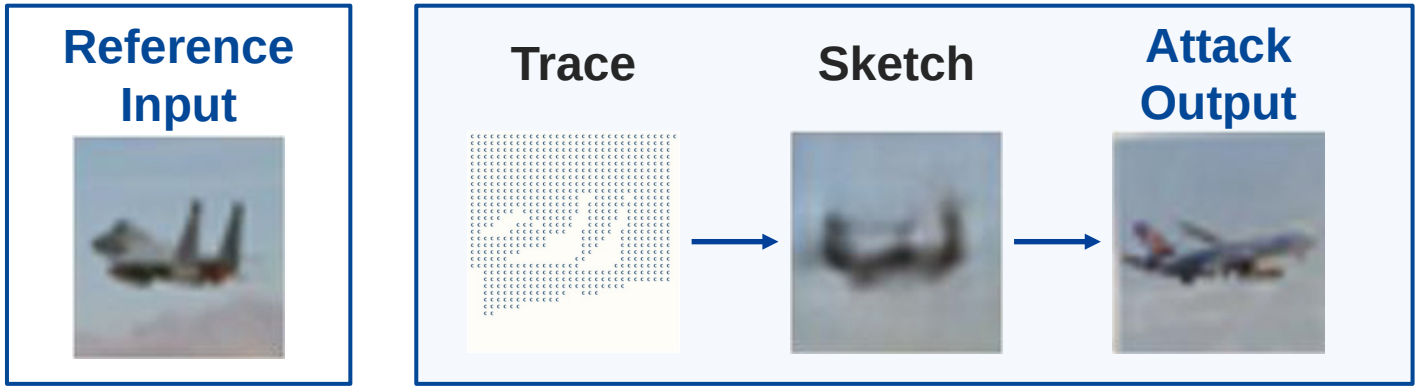
CipherSteal Example: reconstructed image

Attack Output ⇒ Leakage?



Attributing Attack Output

Victim-trace example



Reconstruction = trace part + prior part

Empty-trace example

No victim information



Reconstruction quality is not a valid criterion

Our criterion: exclude the prior part, count only what the victim's trace adds

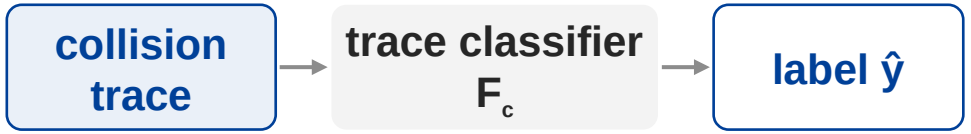
Measuring the Attacker's Gain

Trace utility (TU)



$$TU = score_{victim} - score_{mismatch}$$

Trace-to-label attack



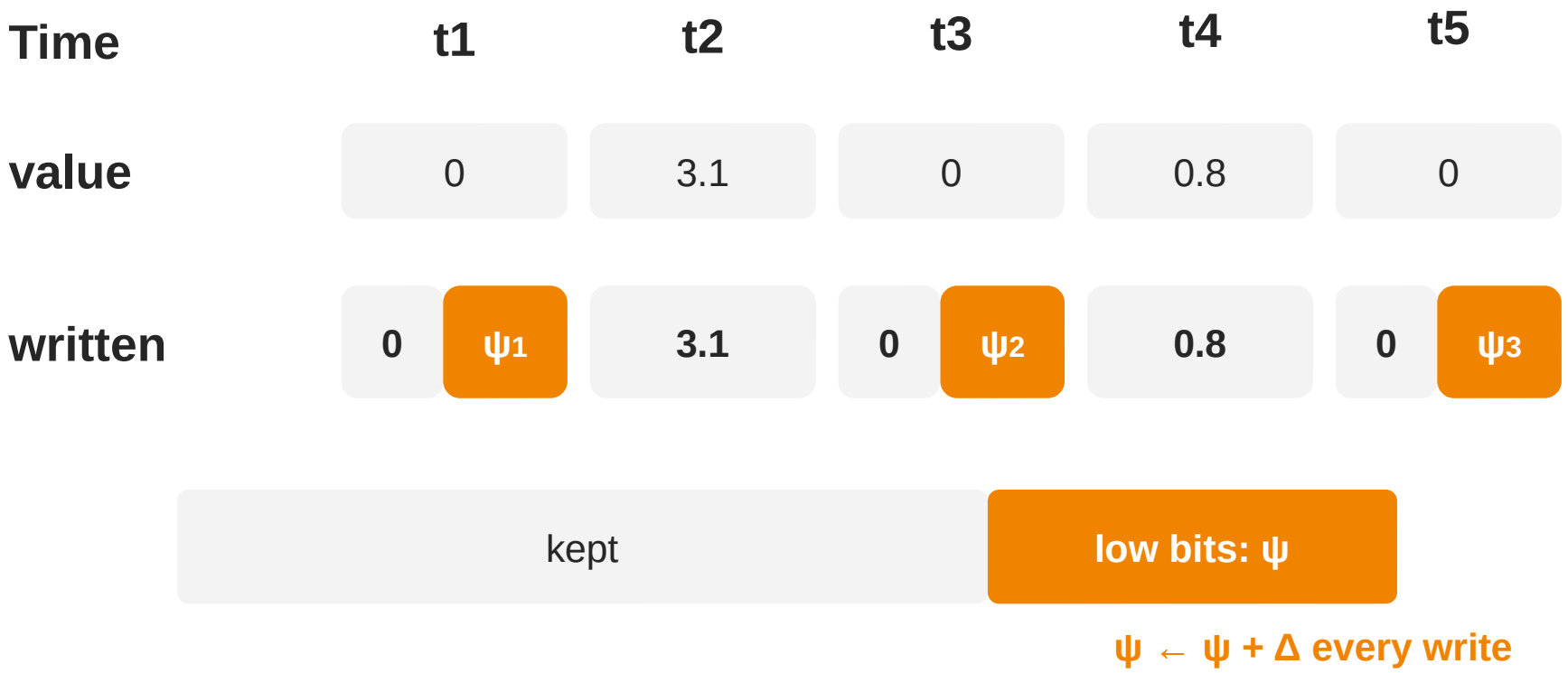
Cons: rate of $\hat{y}$ = victim's prediction

$$TU^{cons} = Cons_{victim} - Cons_{mismatch}$$

Defense goal: $TU \approx 0$

Defense Design

Rewrite the low bits at every write



Trace Utility Results

$$TU^{cons} = Cons_{victim} - Cons_{mismatch}$$

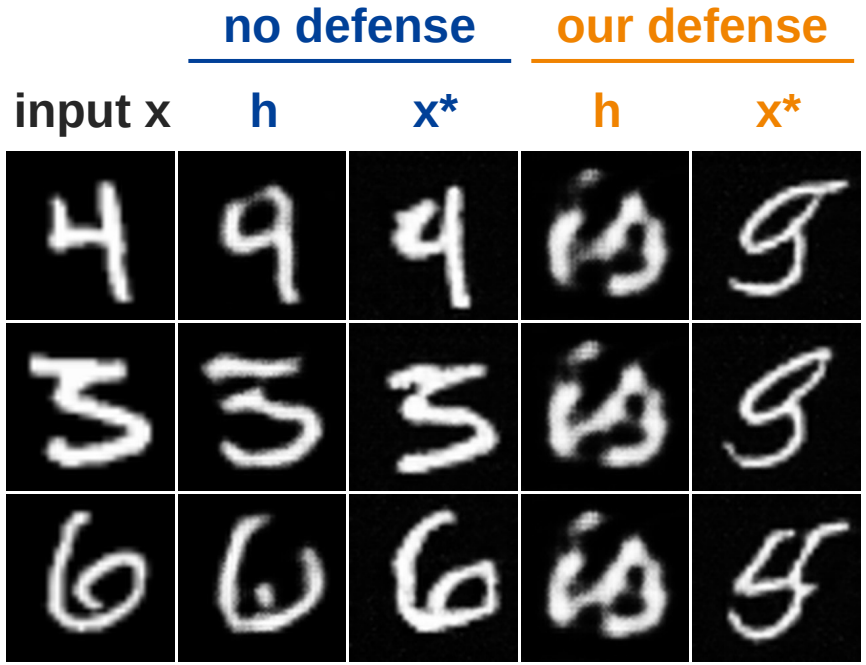
Dataset	Ref. Acc.	Setting	Pred. Acc.	$TU_{\tau}^{cons}(A, K)$
CIFAR	0.7860	Def^{-}	0.7564	0.7652
		Def^{+}	0.1002	0.0001
ImageNet	0.8350	Def^{-}: `transition1.conv`	0.4368	0.4248
		Def^{-}: `transition2.conv`	0.6160	0.6012
		Def^{-}: `transition3.conv`	0.7484	0.7508
		Def^{+}: `transition1.conv`	0.2400	0.2244
		Def^{+}: `transition2.conv`	0.1656	0.1460
		Def^{+}: `transition3.conv`	0.0200	0.0000
MNIST	0.9820	Def^{-}	0.9540	0.8360
		Def^{+}	0.0980	0.0000
Chest X-ray	0.7770	Def^{-}	0.7087	0.1915
		Def^{+}	0.5128	0.0000

Reconstruction Results

CIFAR



MNIST



Consistent with TU ≈ 0: the attack output no longer depends on the input

Further Evaluation

Task Utility

Table 6: Task utility comparison results (with absolute changes from Def$^-$ in parentheses).

Task	Def$^-$	MC (Def$^+$)	IC (Def$^+$)	TV (Def$^+$)	TA (Def$^+$)
T01	98.91%	98.91%(+0.0000)	98.91%(+0.0000)	98.91%(+0.0000)	98.91%(+0.0000)
T02	99.47%	99.47%(+0.0000)	99.47%(+0.0000)	99.47%(+0.0000)	99.47%(+0.0000)
T03	93.98%	93.89%(-0.0009)	93.94%(-0.0004)	93.85%(-0.0013)	93.85%(-0.0013)
T04	92.17%	92.08%(-0.0009)	92.19%(+0.0002)	92.19%(+0.0002)	92.20%(+0.0003)
T05	63.12%	63.12%(+0.0000)	63.12%(+0.0000)	63.08%(-0.0004)	63.12%(+0.0000)
T06	99.54%	99.55%(+0.0001)	99.54%(+0.0000)	99.55%(+0.0001)	99.55%(+0.0001)
T07	94.80%	94.84%(+0.0004)	94.81%(+0.0001)	94.76%(-0.0004)	94.75%(-0.0005)
T08	71.84%	71.80%(-0.0004)	71.84%(+0.0000)	71.92%(+0.0008)	71.80%(-0.0004)
T09	53.32%	53.18%(-0.0014)	53.42%(+0.0010)	53.02%(-0.0030)	52.98%(-0.0034)
T10	77.67%	77.35%(-0.0032)	77.67%(+0.0000)	77.68%(+0.0001)	77.68%(+0.0001)
T11	99.45%	99.47%(+0.0002)	99.45%(+0.0000)	99.45%(+0.0000)	99.45%(+0.0000)
T12	93.98%	93.48%(-0.0050)	94.02%(+0.0004)	94.01%(+0.0003)	94.00%(+0.0002)
T13	59.14%	58.76%(-0.0038)	59.06%(-0.0008)	58.88%(-0.0026)	58.86%(-0.0028)
T14	77.42%	77.44%(+0.0002)	77.44%(+0.0002)	77.46%(+0.0003)	77.46%(+0.0003)
T15	82.76%	82.80%(+0.0004)	82.72%(-0.0004)	82.84%(+0.0008)	82.76%(+0.0000)
T16	73.36%	72.98%(-0.0038)	73.50%(+0.0014)	73.50%(+0.0014)	73.54%(+0.0018)
T17	52.17%	51.34%(-0.0083)	52.18%(+0.0001)	52.18%(+0.0001)	52.18%(+0.0001)

Runtime Overhead

Table 7: Runtime overhead relative to Def$^-$. Each value reports the median over five de-noised runs.

Task	Model/Dataset	MC	IC	TV	TA
T01	LeNet/MNIST	+4.90%	+3.85%	+10.11%	+12.88%
T03	VGGNet/CIFAR	+26.89%	+0.56%	+0.63%	+0.51%
T07	ResNet/CIFAR	+8.50%	+0.74%	+0.67%	+0.64%
T10	ResNet/Chest X-ray	+6.52%	+0.29%	+0.72%	+0.89%
T17	DenseNet/Chest X-ray	+53.96%	+2.87%	+2.87%	+2.81%

Table 8: Comparison of task utility and inference runtime overhead of different defense schemes. Results report test outcomes on selected task groups for three defense schemes (Ours, CIPHERFIX-style, and ZEBRAFIX-style).

Task	Task Utility				Time Overhead		
	Ref. (Def$^-$)	Ours (Def$^+$)	CIPHERFIX-style	ZEBRAFIX-style	Ours (Def$^+$)	CIPHERFIX-style	ZEBRAFIX-style
T01	0.9891	0.9891	0.9891	0.9891	+4.68%	+962.65%	+588.19%
T03	0.9398	0.9397	0.9398	0.9398	+14.69%	+2466.30%	+1943.12%
T07	0.9409	0.9414	0.9409	0.9409	+5.90%	+691.96%	+658.96%

Thank you!